

Bounded Communication Fault Tolerance

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Introduction

Today:

- ▶ Bounded Communication Fault Tolerance
 1. Walkthrough the math
 2. Alternative definition to Expander decomposition
 3. (d, k) - Toric
- ▶ New experimental results.
- ▶ Answering the code teleportation.

Bounded Communication Fault Tolerance

Setting

We allow a non-nosy, strong as we wish, classical computation aside. Yet we would like to limit the communication to a constant number of bits per logical qubit, in each round.

Justification

1. If we don't have a constant overhead in depth, in that model,
 $\Rightarrow$ we also don't have it in the standard model.

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2. In that model, gate teleportation for a constant dimension code is easy. Namely, for the Topological codes the task is reduced to finding a decoding with a constant-size hint.

Max Color

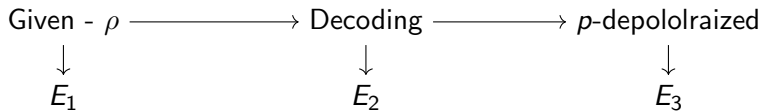
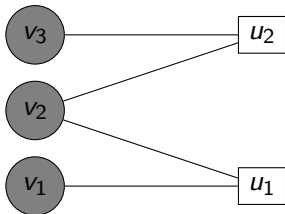
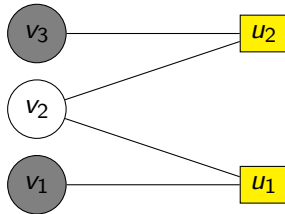


Figure: Illustration of the cycle.

v_2 is flipped.



v_2 is not faulty.



If the majority of bits in the local view of v_2 are flipped, then after decoding, v_2 will be also (remain) flipped.

Strategy

Suppose we have the following Lemma:

Nice To Have Lemma.

There is a function $f : [2^n] \rightarrow \mathbb{R}$, such that for any (encoded) ρ which is subjected to q -local stochastic noise and a subset of qubits S we have:

1. With high probability $(1 - e^{-\sqrt{n}})$, after the decoding stage, the probability that the error is (partly) supported on S is bounded by $C \cdot q^{f(|S|)}$
2. $f(|S|) \geq \alpha|S|$ for some $\alpha > 0$.

Strategy

The Nice-To-Have-Lemma implies that after decoding + absorbing p -depolarized noise, we have with high probability:

$$\Pr[S] \leq \sum_{S' \subset S} C \cdot q^{f(S')} p^{|S/S'|} \leq C \cdot (q^\alpha + p)^{|S|} \leq C \cdot q^{|S|}$$

SWIFT

What did we learn?

1. Storing time is, likely, not $\Theta(e^n)$, might be $\Theta(L^\alpha)$ for $\alpha \in (0, 1)$.
2. $\Rightarrow$ Implies that the structure of the theorem/key-lemma is different than the known refrigerators.

Future

So What's Next?

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 - ▶ More experiments, higher statistical standards?
 - ▶ Accelerate the decoder? Shifting to *c/c++*? Parallelize it with GPUs? Do we have budget?

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3. How to use the fact our code is a quotient code?